

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	TM-RISINGWATERS-04
Project Title	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform flood emergency mitigation and local risk modeling using machine learning, boosting operational response speeds and hazard prediction accuracy. It tackles municipal safety system inefficiencies, promising better field coordinates, reduced life/asset risks, and safer local communities. Key features include a machine learning-based hydrological prediction model and automated real-time emergency alert dispatch.

PROJECT OVERVIEW	
Objective	The primary objective is to revolutionize modern disaster response frameworks by implementing predictive deep learning algorithms to achieve precise, street-level flood hazard mapping.
Scope	The project comprehensively synthesizes local river basin telemetry, municipal topographical GIS maps, and weather satellite data outputs to compute localized inundation heights, generating real-time community safety alerts.
PROBLEM STATEMENT	
Description	Existing warning methods rely on broad statistical estimates without pinpointing street-level risks. This information deficit delays municipal rescue efforts, leads to citizen inaction, and compounds infrastructure damages.
Impact	By establishing hyper-local models, emergency agencies can optimize evacuation patterns, mitigate severe human hazard casualties, protect local assets, and provide structured peace of mind.
PROPOSED SOLUTION	
Approach	Applying robust convolutional neural networks (CNNs) and LSTM networks to process real-time meteorological forecasts, regional elevation grids, and tidal meters to anticipate risk paths up to 24 hours out.
Key Features	• Dynamic street-level hazard inundation modeling. • Automatic regional crisis broadcasting dispatch via targeted Twilio SMS API. • Interactive GIS dashboard built on React and high-performance Leaflet.js.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 Tensor Core GPU (or equivalent cloud compute vGPU), 4-Core Intel Xeon vCPU
Memory	RAM specifications	16 GB System Memory
Storage	Disk space for data, models, and logs	256 GB SSD (local system storage for raster grids) + AWS S3 Cloud Storage allocation
Software		
Frameworks	Python web & machine learning frameworks	FastAPI, PyTorch / TensorFlow
Libraries	Additional support libraries	scikit-learn, pandas, numpy, matplotlib, Leaflet.js, rasterio (for satellite GeoTIFFs)
Development Environment	IDE	VS Code, Jupyter Notebook, GitHub
Data		
Data Source	Origin, volume, and dataset formats	OpenWeatherMap API meteorological streams, National Hydrological Agency river telemetry (.csv), regional GIS elevation maps (.tif)